Future & Promise

Season: 5 · Concurrency · Category: Concurrency · Level: Intermediate

Intent

- Represent a value that will arrive asynchronously.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Order tracking number before delivery.

When to Use

- Concurrent / parallel workloads.
- The design force named in Intent shows up more than once.

When Not to Use

- Pure CPU-bound work already parallelized.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.concurrency.futurepromise;

import java.util.concurrent.CompletableFuture;
import java.util.concurrent.CompletionStage;
import java.util.concurrent.Future;

/** Promise/future demo using CompletableFuture (Java's standard promise). */
public final class Promise {
    private Promise() {}

    public static CompletionStage<Integer> computeAsync(int x) {
        return CompletableFuture.supplyAsync(() -> x * 2);
    }

    public static Future<Integer> asFuture(int x) {
        return computeAsync(x).toCompletableFuture();
    }
}
```

Modern Java 25 / Functional

- Executors.newVirtualThreadPerTaskExecutor() is the modern default for blocking tasks.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: concurrency/futurepromise/
```

Pitfalls

- Prefer java.util.concurrent and virtual threads over custom pools.
- Always interrupt/shutdown cleanly.

Interview Q&A

- Q: Virtual threads vs platform threads? A: Virtual threads excel at blocking I/O concurrency; still bound by CPU for pure compute.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Represent a value that will arrive asynchronously.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Order tracking number before delivery. [0:40] Intent — State the design force in one sentence: Repre...